

OLMo ICL / Semantic Induction Head Development

Two-page summary of the completed developmental side experiment – September 24, 2026

Research question and experimental design

Question. Does OLMo-2-1124-7B acquire few-shot in-context learning (ICL) during the same training phase in which Semantic Induction Head (SIH)-like relational structure emerges?

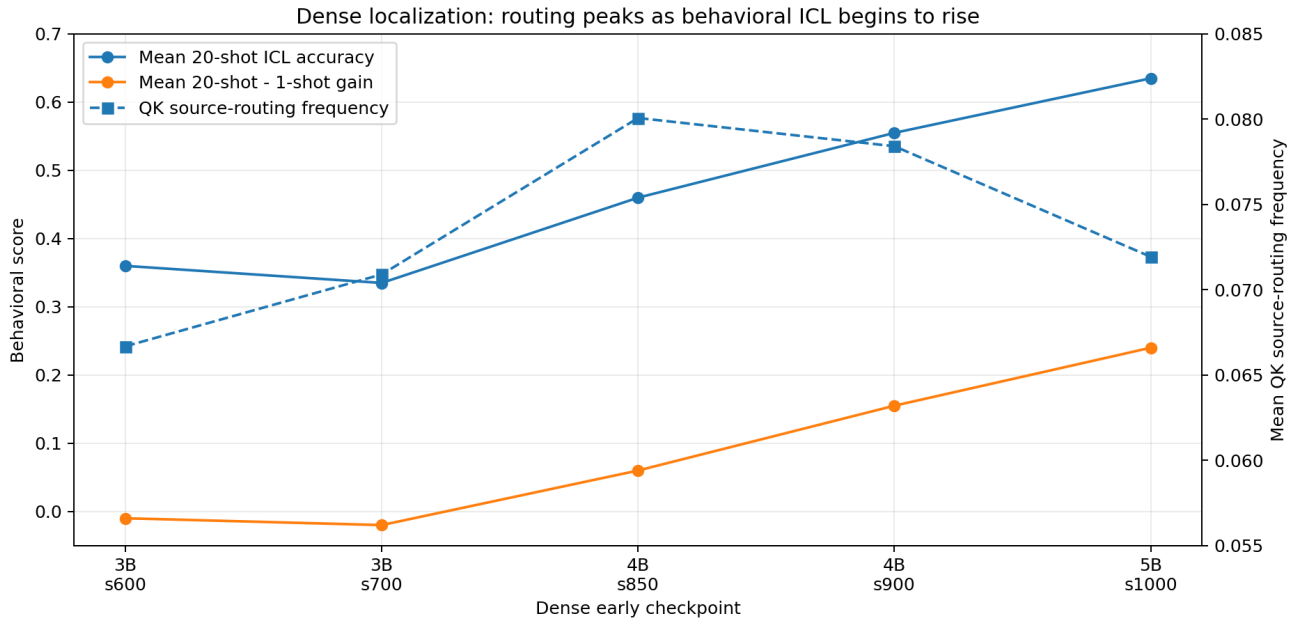
Behavioral ICL. We evaluated four synthetic classification tasks (binary fruit/month, binary furniture/profession, four-class, nine-class) at 0, 1, 2, 3, 4, 5, 10, and 20 shots, with 50 frozen examples per task-shot condition. The same prompts were reused at every checkpoint. Seventeen Stage-1 checkpoints were analyzed; the dense early set is steps 600, 700, 850, 900, and 1000 (approximately 3B, 3B, 4B, 4B, and 5B cumulative training tokens).

Developmental RI. We evaluated all 1,024 attention heads on a balanced 700-triplet AGENDA semantic-relation set (100 examples for each of seven relations). Following Ren et al.'s developmental procedure, the primary QK gate requires the source token to be the attention argmax; the original $\tau = 2.2$ dominance gate is retained as a sensitivity check. Conditional Relation Index (RI) measures OV tail-token promotion only after source routing occurs. We also record QK source-routing frequency separately so that routing and conditional OV selectivity are not conflated.

Behavioral ICL localizes near 4B tokens

Step	Approx. tokens	Mean 20-shot acc.	Mean (20 – 1) gain	Tasks CI > chance
600	3B	0.360	-0.010	0 / 4
700	3B	0.335	-0.020	0 / 4
850	4B	0.460	+0.060	2 / 4
900	4B	0.555	+0.155	4 / 4
1000	5B	0.635	+0.240	4 / 4

Key result. The behavioral transition begins between steps 700 and 850 and is reliable across all four tasks by step 900. The first sampled checkpoint at which every 20-shot Wilson interval is above chance is step 900, so the best current localization is **approximately 4B training tokens**.



Format precedes pattern discovery. Mean 20-shot format accuracy is already 1.00 at steps 600 and 700 and stays there through the transition, while mapping accuracy is still weak. This reproduces the qualitative ordering emphasized by Ren et al.: surface-format compliance develops before reliable pattern discovery.

Mechanistic result: routing changes, conditional RI does not

The dense RI trajectory reveals a transient early increase in source-directed QK routing that peaks at step 850, directly adjacent to the behavioral transition. Relation-balanced QK frequency rises from 0.0667 at step 600 to 0.0801 at step 850 and remains elevated at 0.0784 at step 900. Conditional RI, however, stays near 0.08 throughout this interval.

Comparison	Δ QK	95% paired-cluster CI	Fixed-support Δ RI (95% Bayes interval)
600 $\rightarrow$ 850	+0.01339	[0.01010, 0.01633]	-0.000326 [-0.001176, 0.000452]
600 $\rightarrow$ 900	+0.01173	[0.00801, 0.01517]	-0.000161 [-0.001094, 0.000693]
600 $\rightarrow$ 1000	+0.00525	[0.00077, 0.00942]	+0.000951 [-0.000129, 0.001853]

The fixed-support analysis keeps only relation-head cells with at least 10 scored source-routing events at both endpoints and holds that population fixed across paired Bayesian cluster-bootstrap replicates. Dirichlet chunk weights eliminate the zero-denominator problem, giving a zero-denominator rate of exactly 0 in all 5,000 replicates. All three fixed-support RI intervals include zero.

Key result. The robust developmental event is **QK/source-routing reorganization**. Within the same adequately supported heads, conditional OV tail selectivity is approximately stable through the step-850/900 ICL transition. The early RI-per-opportunity peak is therefore driven primarily by routing frequency, not by a population-wide jump in conditional RI.

Head identity and relation to prior experiments

The mature high-RI head population is not already stable at ICL onset. Mean top-15 Jaccard overlap with the final checkpoint is 0.000 at steps 600 and 850 and only about 0.005 at steps 900 and 1000; all-head rank similarity to the final model is also low in this window. Final high-RI heads acquire much stronger semantic target preference only later in training.

This remains consistent with the earlier single-hop mechanistic experiments: mature-model RI was a weak causal head-ranking metric for the synthetic mother-of task. The developmental result therefore does not rescue RI as a causal selector; it instead suggests that an early routing reorganization can temporally accompany ICL acquisition while mature RI-style specialization continues to reorganize later.

Comparison with Ren et al. and final conclusion

Agreement. OLMo’s behavioral pattern-discovery transition occurs at roughly the same token scale as Ren et al.’s reported $\sim$ 4B-token regime, and format compliance precedes pattern discovery in both settings. The developmental QK gate and OV-based RI logic are also closely matched.

Difference. We do *not* observe a broad early increase in conditional RI itself. The strongest aligned internal signal is QK routing. In addition, OLMo shows substantial later head-population reorganization and a late decline in toy-task ICL accuracy despite preserved format compliance.

Key result. Final conclusion: OLMo exhibits a temporally aligned early relational-routing reorganization that peaks at step 850, just as behavioral ICL begins to rise and shortly before it is reliable across all tasks at step 900. This is **partial support** for the SIH/ICL developmental picture, but not a direct replication of a population-wide conditional-RI emergence. The current evidence supports temporal association, not causality.

Design caveat. The broad step-600–1000 behavioral window was established before interpreting RI timing, but steps 700/850/900 were added after the RI peak motivated closer inspection. The exact 850/900 match is therefore targeted follow-up evidence rather than an independently preregistered temporal prediction.

Stopping point. The main robustness checks are complete: dense behavioral localization, paired QK bootstrap, strict-gate sensitivity, null controls, head-stability analysis, and fixed-support Bayesian RI sensitivity. Further work (paper-fidelity preprocessing/reverse relations or causal ablation/patching of early routing heads) would be an extension rather than required validation.